

ETF Portfolio Analysis

ECON 424 - Group 8

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Agenda:

- Asset Selection
- Key Findings/Executive Summary (Includes findings and lessons from the manipulation of the data of the assets to avoid presenting everything and keep presentation concise and focused on key messages)
- Further detail into the key lessons and findings
- Investment Implications

Asset Selection:

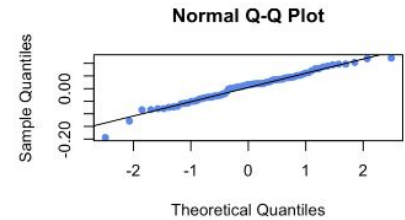
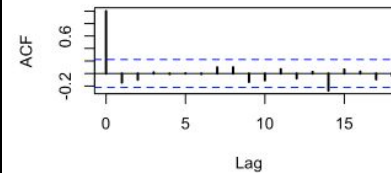
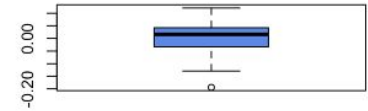
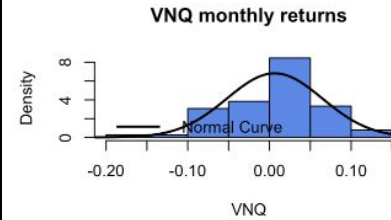
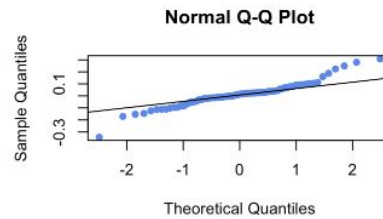
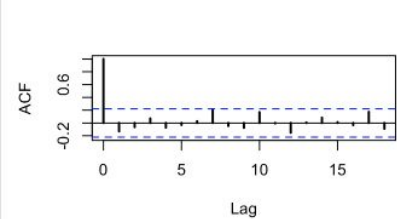
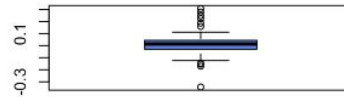
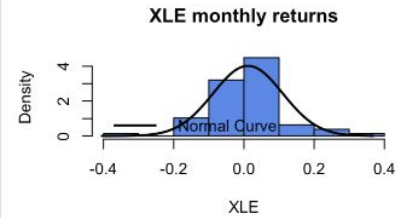
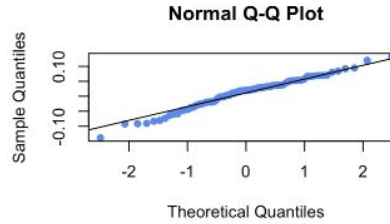
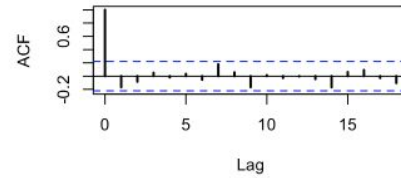
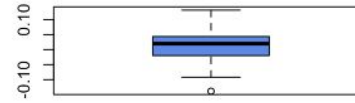
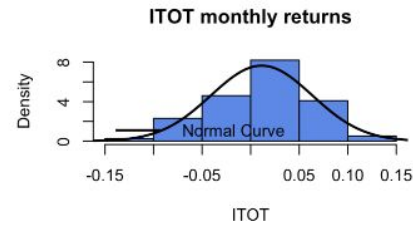
Our ETF Selection

We selected a diverse mix representing different market segments:

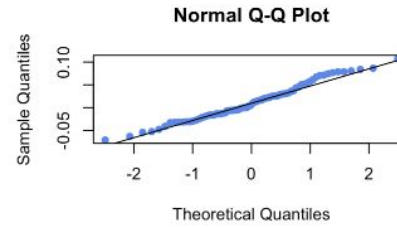
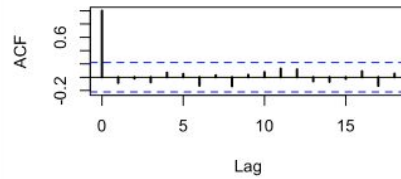
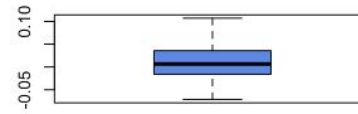
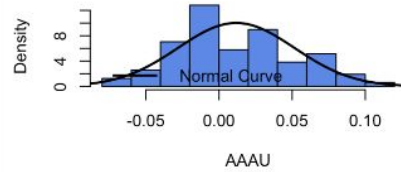
- **ITOT**: iShares Core S&P Total U.S. Stock Market ETF
- **VNQ**: Vanguard Real Estate ETF
- **XLE**: Energy Select Sector SPDR
- **QQQ**: Invesco QQQ Trust (Nasdaq-100/Tech)
- **IWM**: iShares Russell 2000 ETF (Small Caps)
- **AAAU**: Goldman Sachs Physical Gold ETF

Analysis Period: ~7 years (Sept 2018- March 2025)

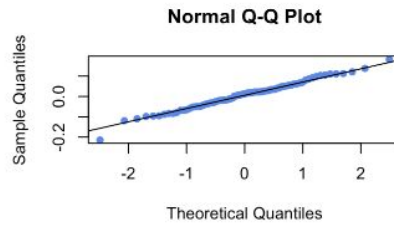
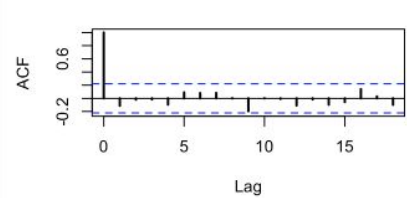
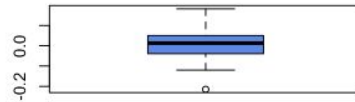
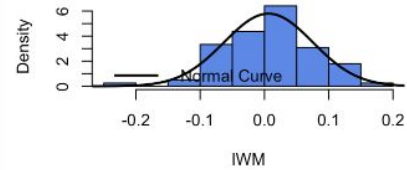
Asset returns:



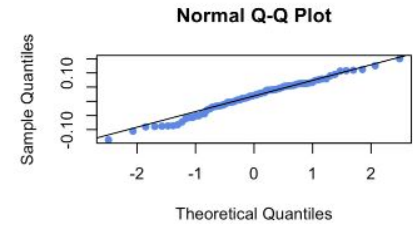
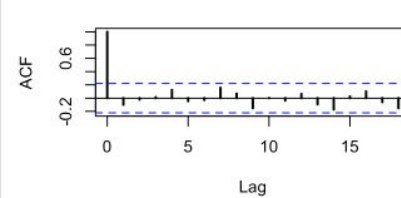
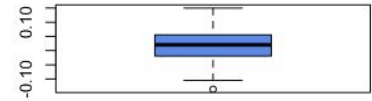
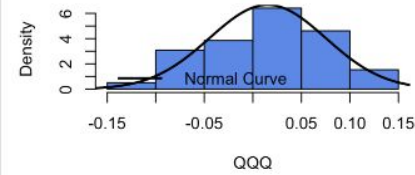
AAAU monthly returns



IWM monthly returns

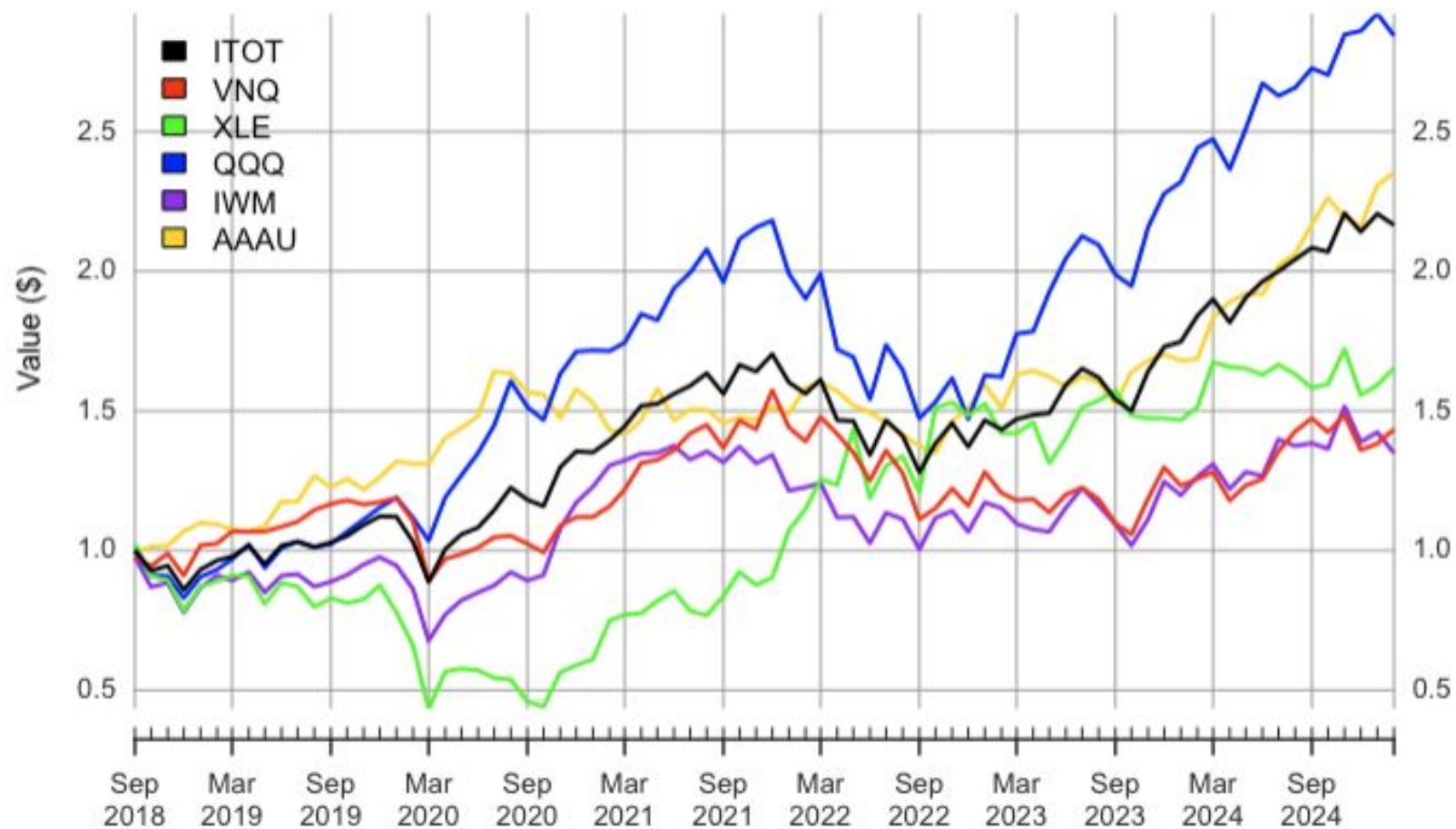


QQQ monthly returns

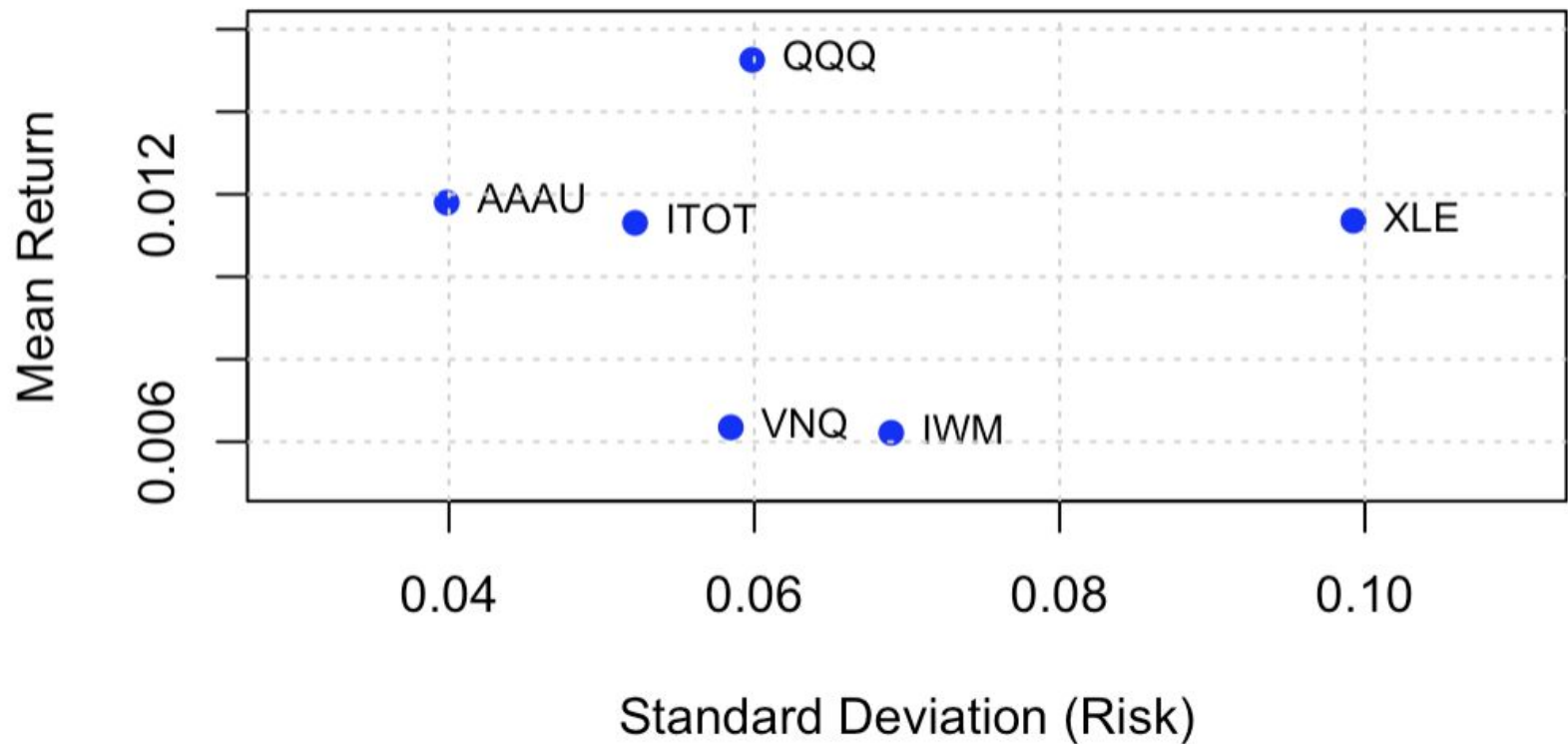


Growth of \$1 Investment (2018-2025)

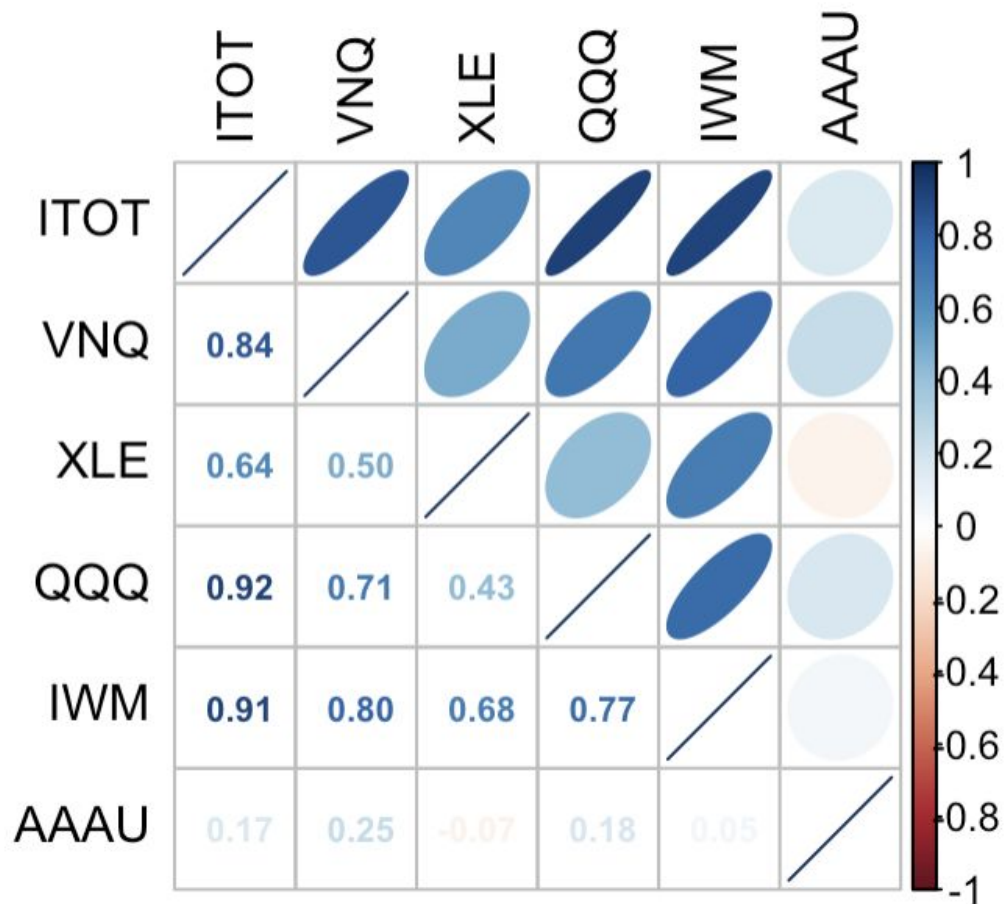
Sep 2018 / Feb 2025



Risk-Return Relationship



Correlation Matrix of Asset Returns



Key Findings:

Executive Summary

- **Gold's Unexpected Value:** AAAU delivered strong returns (14.16% annually) with the lowest volatility (13.83%), providing exceptional diversification with negative correlation to energy sector
- **Optimized Portfolios Outperform:** Global Minimum Variance Portfolio achieved both higher returns (14.2% vs 12.5%) and lower risk (11.5% vs 17.4%) than equal-weighted allocation
- **Risk Concentration Matters:** In an equal-weighted portfolio, XLE (Energy) contributes 35.2% of portfolio risk despite only 16.7% of capital
- **Sector Performance Divergence:** QQQ (Tech) led with 18.3% annual returns while Real Estate (VNQ) and Small Caps (IWM) significantly underperformed at ~7.5% annually
- **Benefits of Being Picky:** A 15% target return achieved most efficiently with 71.2% QQQ and 28.8% AAAU, demonstrating the power of selective asset allocation

The Power of Diversification

Critical Correlation Findings:

- AAAU (Gold) negatively correlated with XLE (Energy) (-0.065)
- AAAU minimally correlated with all equities (0.05-0.25)
- High correlation among equity ETFs (0.71-0.92)

Value-at-Risk Comparison (1% VaR on \$100K investment):

- XLE (Energy): \$21,952 (highest risk)
- AAAU (Gold): \$8,091 (lowest risk)
- GMVP Portfolio: \$7,724 (below any individual asset)

Key Insight: Combining assets with low/negative correlations dramatically reduces portfolio risk

Portfolio Optimization Benefits

Portfolio	Annual Return	Annual Risk	Sharpe Ratio
Equal-Weight	12.5%	17.4%	0.688
GMVP	14.2%	11.5%	1.190
Tangency	22.3%	14.9%	1.370

Global Minimum Variance Portfolio (GMVP) Composition:

- ITOT: 111.3%
- AAAU: 62.4%
- Negative weights in VNQ, XLE, QQQ, and IWM

Key Insight: Optimization creates the counterintuitive result of higher returns with lower risk

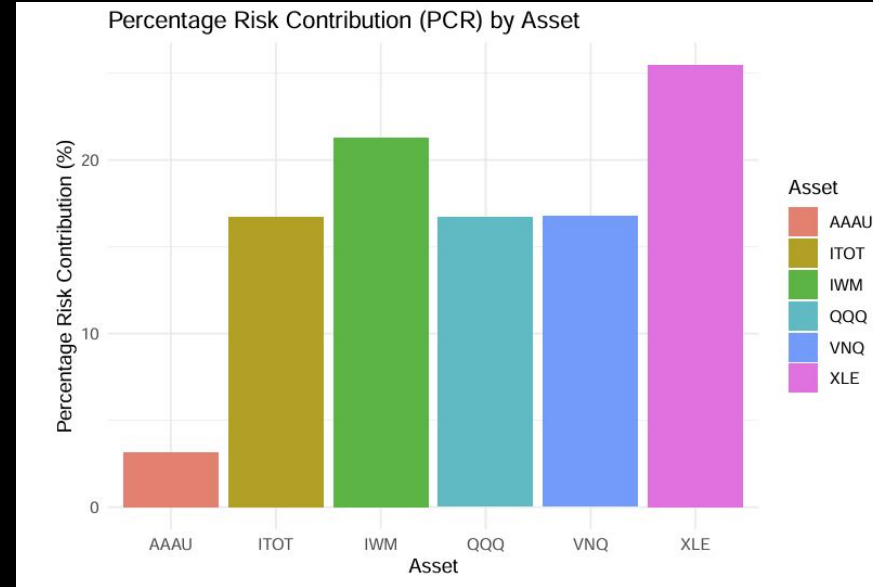
Equal Allocation $\neq$ Equal Risk

Equal-Weighted Portfolio Risk

Contribution(16.67% weighted for each asset):

- XLE (Energy): 25.47% of risk
- IWM (Small Caps): 21.26% of risk
- AAAU (Gold): 3.13% of risk
- ITOT: 16.72% of risk
- VNQ: 16.75% of risk
- QQQ (Tech): 16.68% of risk

Key Takeaway: The percent of risk attribution does not match the percent of asset allocation. In other words, equal allocation does not mean equal risk. XLE contributes the most, while AAAU contributes the least to the portfolio risk.



Performance & Risk Assessment

Asset	Annual Return	Annual StdDev	Sharpe Ratio	\$1 Growth (10yr)
QQQ	18.31%	20.68%	0.859	\$2.84
AAAU	14.16%	13.83%	0.989	\$2.31
XLE	13.65%	34.35%	0.382	\$2.10
ITOT	13.56%	18.86%	0.723	\$2.05
VNQ	7.52%	19.93%	0.351	\$1.42
IWM	7.46%	23.91%	0.291	\$1.39

Key Insight:

Tech's dominance (QQQ 18.31%) reflects the digital transformation driving economic growth while gold's strong performance (AAAU 14.16%) demonstrates its evolving role beyond inflation hedging. Meanwhile, real estate and small caps underperformed amid rising interest rates, while energy's volatility reflects ongoing sector transition challenges (renewable energy competition and supply disruptions).

Efficient Target Portfolios

1.25% Target Monthly(15% Annual) Return Asset Allocation (6 ETFs and No Short Sales):

- ITOT: 2.0%
- VNQ: 0.0% (Low annual Return)
- XLE: 74%
- QQQ (Tech): 21.5% (Highest annual Return)
- IWM : 0.0% (Lowest annual Return)
- AAAU (Gold): 69.2% (High annual Return)

Portfolio Characteristics based on \$100000 investment:

- Monthly StdDev: 3.47%
- 1% Monthly VaR: \$8087.47
- 5% Monthly VaR: \$5709.82

Key Takeaway: This portfolio has moderate risk with relatively high exposure to XLE, QQQ, and AAAU, contributing to volatility. Investors should be comfortable with fluctuations of ~\$3,470 per month on average and prepared for worst-case monthly losses exceeding \$8,087 (1% VaR).

0.5% Target Monthly(6% Annual) Return Asset Allocation (6 ETFs and Risk free asset with a monthly return 0.167% and No Short Sales):

Weight in Risk-Free Asset: 61.77%

Weight in Risky Portfolio: 38.23%

- ITOT: 1.82%
- VNQ: 3.55% (Low annual Return)
- XLE: 0.77%
- QQQ (Tech): 0.0% (Highest annual Return)
- IWM : 6.02% (Lowest annual Return)
- AAU (Gold): 26.07% (High annual Return)

Key Takeaway: This portfolios aims for relatively low annual returns, so it is mainly composed of low-risk and risk-free assets with low annual return when there is no shorts allowed.

Equally Weighted Portfolio

Asset	Annual Return	Annual Std Dev	Annual Sharpe Ratio
ITOT	0.136	0.181	0.723
VNQ	0.076	0.202	0.352
XLE	0.136	0.344	0.382
QQQ	0.183	0.207	0.859
IWM	0.075	0.239	0.291
AAAU	0.142	0.138	0.989

Equally Weighted Portfolio:
Annualized Expected Return: 0.125
Annualized Volatility: 0.174
Annualized Sharpe Ratio: 0.688

The volatility is slightly higher than AAAU (13.8%), which is the least volatile asset. meaning the portfolio benefits from risk diversification. This shows that spreading investment across different asset classes reduces individual asset risks.

The annualized Sharpe Ratio (0.688) is higher than XLE (0.382), VNQ (0.351), and IWM (0.291). It's lower than AAAU (0.989), ITOT(0.723) and QQQ (0.859).

Global Minimum Variance Portfolio

Portfolio Weights:

ITOT	VNQ	XLE	QQQ	IWM	AAAU
1.1130	-0.1680	-0.0202	-0.3793	-0.1695	0.6240

Expected Monthly Return: 0.0118

Monthly Standard Deviation: 0.0332

The volatility (3.32%) is significantly lower than that of any individual asset, demonstrating the benefits of diversification.

Gold (AAAU) and Total Market (ITOT) are the dominant holdings, making up 62.4% and 111.3% of the portfolio, respectively. The GMVP includes negative weights for VNQ, XLE, QQQ, and IWM, indicating short-selling in these assets.

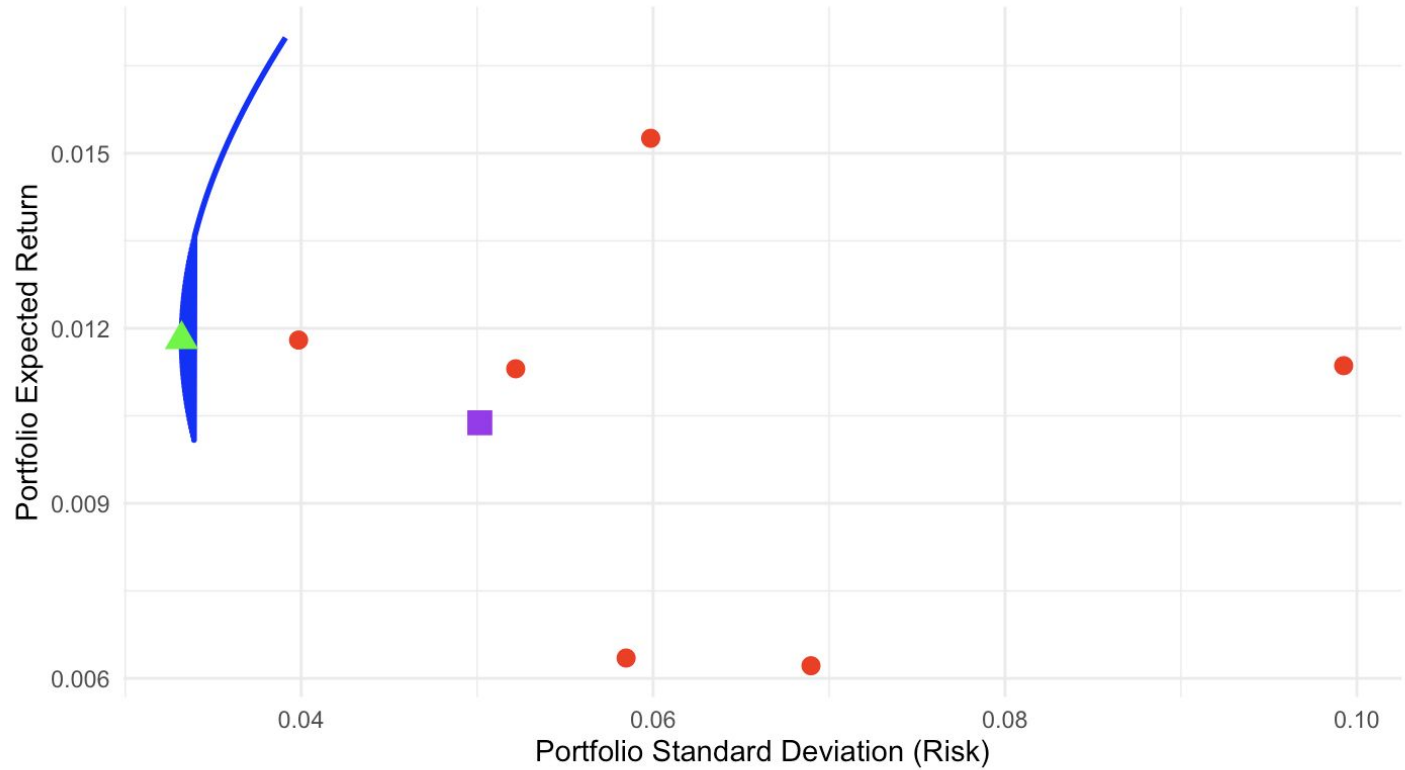
Annualized GMVP Results:

- Annualized Expected Return: 0.142
- Annualized Volatility: 0.115
- Annualized Sharpe Ratio: 1.06

GMVP (11.5%) has the lowest volatility among all individual assets and the equally weighted portfolio (17.4%). GMVP return (14.2%) is higher than the equally weighted portfolio (12.5%). It's lower than QQQ (18.3%) but offers far less risk.

The GMVP is a safe portfolio with the lowest volatility while maintaining a strong return. It has a good risk-adjusted return (Sharpe ratio = 1.06).

Efficient Frontier with Global Minimum Variance Portfolio



Red = ETFs | Green = GMVP | Purple = Equally Weighted Portfolio

Tangency Portfolio

Asset	Expected Return	Standard Deviation	Sharpe Ratio
Tangency Portfolio	0.019	0.043	0.395
ITOT	0.011	0.052	0.185
VNQ	0.006	0.059	0.080
XLE	0.011	0.099	0.098
QQQ	0.015	0.060	0.227
IWM	0.006	0.069	0.066
AAAU	0.012	0.040	0.254

Annualized Tangency Portfolio Results:

Annualized Expected Return: 0.223

Annualized Standard Deviation: 0.149

Annualized Sharpe Ratio: 1.37

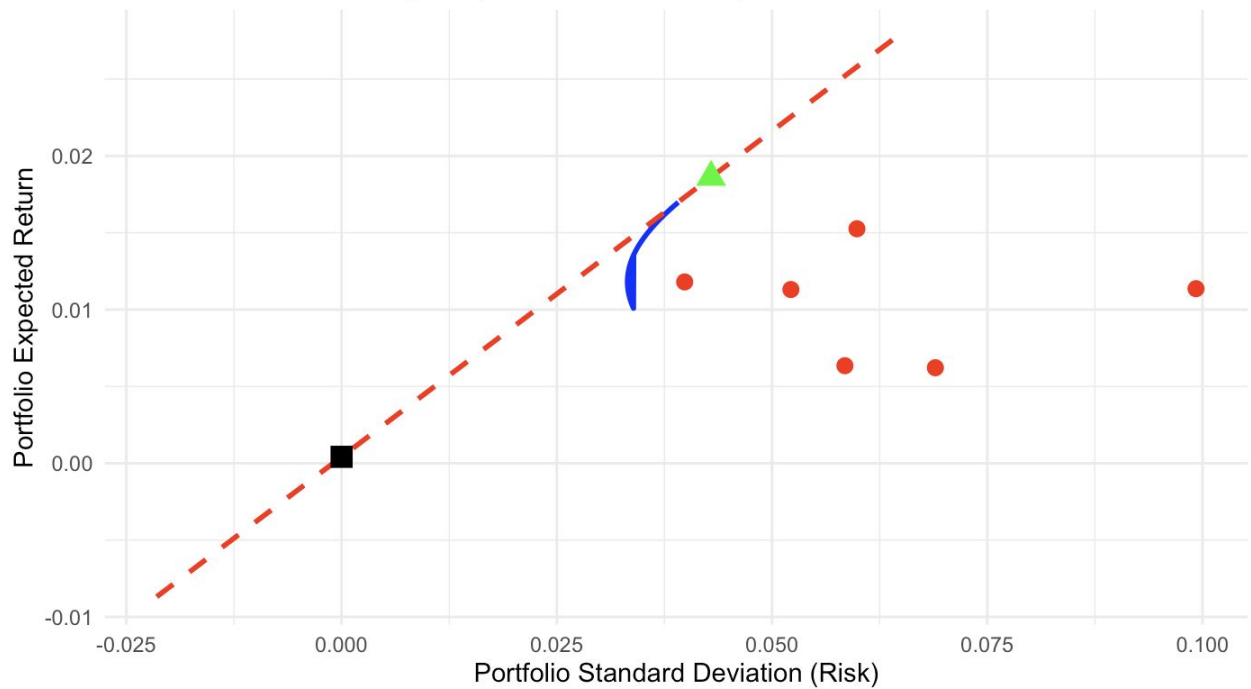
Tangency Portfolio Weights:

ITOT	VNQ	XLE	QQQ	IWM	AAAU
1.059	-0.459	0.147	0.309	-0.687	0.631

VNQ (-45.9%) and IWM (-68.7%) are being shorted. ITOT (105.9%) and AAAU (63.1%) hold large positive weights.

The Tangency Portfolio delivers a high annualized return. The volatility is reasonable. The risk is lower than many individual assets (like XLE, which had high volatility). The Tangency Portfolio offers the best return per unit of risk (1.37 Sharpe Ratio). Higher than all individual assets and other portfolio strategies, confirming its optimal efficiency.

Efficient Frontier, Tangency Portfolio, and Capital Market Line



Blue = Efficient Frontier | Red Dashed = CML | Green = Tangency Portfolio | Black = Risk-Free Asset

GMVP without Short-Sales

Portfolio Weights:

ITOT	VNQ	XLE	QQQ	IWM	AAAU
0.2966	0.0000	0.0345	0.0000	0.0000	0.6688

Gold(66.88%) has strong risk-reducing properties. The broad U.S. stock market (ITOT-29.66%) is still included but at a lower weight than gold. VNQ, QQQ, and IWM were completely excluded, indicating they are too volatile for a low-risk portfolio. XLE (3.45%) were not entirely removed but kept at a very low allocation.

Annualized GMVP (With Short-Selling):

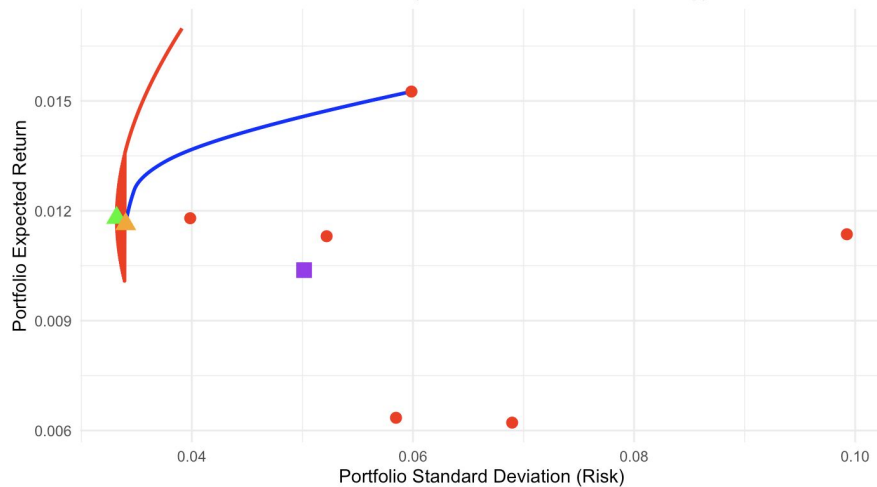
- Annualized Expected Return: 0.142
- Annualized Volatility: 0.115
- Annualized Sharpe Ratio: 1.06

Annualized GMVP (No Short-Selling):

- Annualized Expected Return: 0.14
- Annualized Volatility: 0.118
- Annualized Sharpe Ratio: 1.02

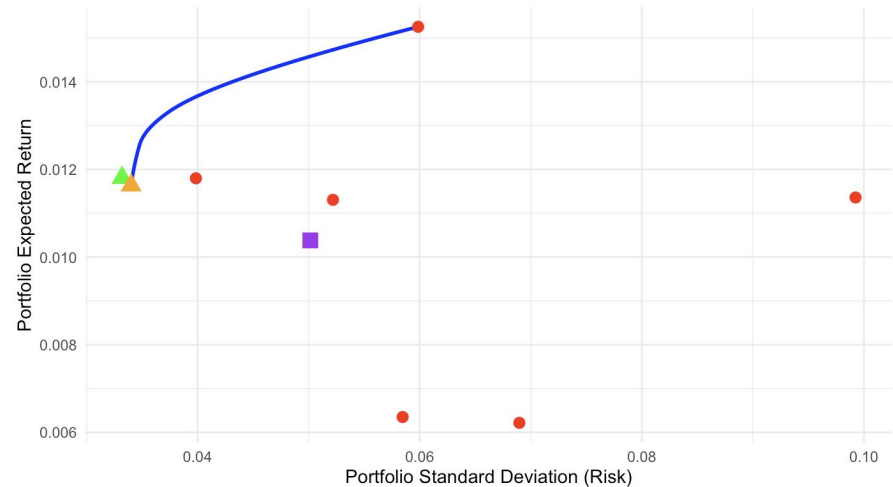
The No Short-Selling GMVP (11.8%) has slightly higher risk than the Short-Selling GMVP (11.5%). No shorts GMVP has 13.9% annualized return and with shorts GMVP has 14.2% annualized return. GMVP (With Shorts) has a slightly better Sharpe Ratio (1.06) than the No Shorts GMVP (1.02).

Comparison of Efficient Frontiers (With & Without Short-Selling)



Red = Short-Selling Allowed | Blue = No Short-Selling | Green = GMVP (Shorts) | Orange = GMVP (No Shorts) | Red Dots = ETFs

Efficient Frontier (No Short-Selling) with GMVP & ETFs



Blue = Efficient Frontier (No Shorts) | Green = GMVP (Shorts Allowed) | Orange = GMVP (No Shorts) | Red = ETFs | Purple = Equally Weighted Portfolio

Tangency Portfolio without Short-Sales

Portfolio Weights (No Short-Selling):

ITOT	VNQ	XLE	QQQ	IWM	AAAU
0.0000	0.0000	0.0447	0.3178	0.0000	0.6375

Gold (AAAU) dominates the portfolio (63.75%), likely due to its low correlation with other assets. Tech stocks (QQQ) hold the second-largest allocation (31.78%). No allocation to VNQ, ITOT, or IWM indicates that they didn't contribute positively in this optimization.

Annualized Tangency Portfolio (With Short-Selling):

- Annualized Expected Return: 0.223
- Annualized Standard Deviation: 0.149
- Annualized Sharpe Ratio: 1.37

Annualized Tangency Portfolio (No Short-Selling):

- Annualized Expected Return: 0.155
- Annualized Standard Deviation: 0.123
- Annualized Sharpe Ratio: 1.09

Value-at-Risk

\$100,000 investment over a one month investment horizon

Asset	1% VaR	5% VaR
GMVP	7724	5461
ITOT	12142	8585
VNQ	13601	9617
XLE	23088	16325
QQQ	13926	9846
IWM	16045	11345
AAAU	9271	6555

GMVP has the lowest risk among all assets. Individual stocks, especially XLE and IWM, have significantly higher tail risk, making them more vulnerable in downturns.

Portfolio	1% VaR	5% VaR
GMVP (No Shorts)	7913	5595
GMVP (With Shorts)	7724	5461

GMVP (No Shorts) has a slightly higher VaR than the Short-Selling GMVP. This indicates that allowing short-selling helps marginally reduce downside risk.

Investment Implications

1. **Gold Deserves Attention:** AAAU's combination of strong returns, low volatility, and diversification benefits challenges conventional wisdom about gold's role in portfolios
2. **Optimization Matters:** Equal-weighted portfolios leave substantial performance on the table - optimization can simultaneously increase returns and reduce risk
3. **Risk Budgeting Is Critical:** Understanding and managing risk contribution is as important as asset allocation
4. **Selective Concentration Works:** For investors seeking 15% returns, a focused allocation to QQQ and AAAU outperforms diversification across all six assets
5. **Tech Leadership Continues:** QQQ's strong performance demonstrates the ongoing technological transformation of the economy

Q&A